

ELITE IGCSE MATHEMATICS
EXPERIENCE NOTES

Sine, Cosine Rule & Area of Triangles

Learn the trigger, write the first line, then solve one similar problem while the method is still fresh.

SINE, COSINE RULE & AREA OF TRIANGLES · CH4 · L11

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The Map

This lesson collapses Sine, Cosine Rule & Area of Triangles into the few exam moves that keep repeating. Read the trigger, copy the first line, then practise the same idea on the next page.

01 Name the angle relationship

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

02 Use the polygon sum

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

03 Use exterior angles

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

04 Chain reasons in order

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

05 Use trigonometry

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

06 Use cosine rule

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BANK EVIDENCE 24 website questions, 24 checked solutions, 3 local PDFs read.

01 Name the angle relationship

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Sine, Cosine Rule & Area of Triangles question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$\text{area} = 9.3 \times 14.7 \times \sin 106^\circ = 131.414 \dots$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P1H Q14

(a)

$$\text{area} = 9.3 \times 14.7 \times \sin 106^\circ = 131.414 \dots$$

Answer

This is a reasoning step: write one clear sentence, then continue the calculation.

Finish: 19.4 cm.

FORMULA BANK

$$\frac{a}{\sin A} = \frac{b}{\sin B}, a^2 = b^2 + c^2 - 2bc \cos A, \text{area} = \frac{1}{2}ab \sin C.$$

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 01

SAME IDEA Use the same first line from Strategy 01.

QUESTION

| In triangle ABC , $AB = 9.4$ cm, $AC = 7.1$ cm and angle $A = 48^\circ$. Find BC .

YOUR SOLUTION

02 Use the polygon sum

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Sine, Cosine Rule & Area of Triangles question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$BC = 13 \sin 40^\circ$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P1HR Q21

Use trigonometry

This is a reasoning step: write one clear sentence, then continue the calculation.

Find the gradient

$$BC = 13 \sin 40^\circ$$

Finish: 12.4 cm

FORMULA BANK

$$\frac{a}{\sin A} = \frac{b}{\sin B}, a^2 = b^2 + c^2 - 2bc \cos A, \text{ area} = \frac{1}{2}ab \sin C.$$

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 02

SAME IDEA Use the same first line from Strategy 02.

QUESTION

| In triangle ABC , $AB = 9.4$ cm, $AC = 7.1$ cm and angle $A = 48^\circ$. Find BC .

YOUR SOLUTION

03 Use exterior angles

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Sine, Cosine Rule & Area of Triangles question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$\text{area} = 9.3 \times 14.7 \times \sin 106^\circ = 131.414 \dots$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P1H Q14

(a)

$$\text{area} = 9.3 \times 14.7 \times \sin 106^\circ = 131.414 \dots$$

Answer

This is a reasoning step: write one clear sentence, then continue the calculation.

Finish: 19.4 cm.

FORMULA BANK

$$\frac{a}{\sin A} = \frac{b}{\sin B}, a^2 = b^2 + c^2 - 2bc \cos A, \text{ area} = \frac{1}{2}ab \sin C.$$

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 03

SAME IDEA Use the same first line from Strategy 03.

QUESTION

| In triangle ABC , $AB = 9.4$ cm, $AC = 7.1$ cm and angle $A = 48^\circ$. Find BC .

YOUR SOLUTION

04 Chain reasons in order

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Sine, Cosine Rule & Area of Triangles question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$\frac{BC}{2} = 14$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2022 P1H Q9

Use cosine rule

$$\frac{BC}{2} = 14$$

Finish: 147 cm².

FORMULA BANK

$$\frac{a}{\sin A} = \frac{b}{\sin B}, a^2 = b^2 + c^2 - 2bc \cos A, \text{ area} = \frac{1}{2}ab \sin C.$$

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 04

SAME IDEA Use the same first line from Strategy 04.

QUESTION

| In triangle ABC , $AB = 9.4$ cm, $AC = 7.1$ cm and angle $A = 48^\circ$. Find BC .

YOUR SOLUTION

05 Use trigonometry

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Sine, Cosine Rule & Area of Triangles question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$\text{area} = 9.3 \times 14.7 \times \sin 106^\circ = 131.414 \dots$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P1H Q14

(a)

$$\text{area} = 9.3 \times 14.7 \times \sin 106^\circ = 131.414 \dots$$

Answer

This is a reasoning step: write one clear sentence, then continue the calculation.

Finish: 19.4 cm.

FORMULA BANK

$$\frac{a}{\sin A} = \frac{b}{\sin B}, a^2 = b^2 + c^2 - 2bc \cos A, \text{ area} = \frac{1}{2}ab \sin C.$$

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 05

SAME IDEA Use the same first line from Strategy 05.

QUESTION

| In triangle ABC , $AB = 9.4$ cm, $AC = 7.1$ cm and angle $A = 48^\circ$. Find BC .

YOUR SOLUTION

06 Use cosine rule

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Sine, Cosine Rule & Area of Triangles question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$\frac{BC}{2} = 14$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2022 P1H Q9

Use cosine rule

$$\frac{BC}{2} = 14$$

Finish: 147 cm².

FORMULA BANK

$$\frac{a}{\sin A} = \frac{b}{\sin B}, a^2 = b^2 + c^2 - 2bc \cos A, \text{ area} = \frac{1}{2}ab \sin C.$$

BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 06

SAME IDEA Use the same first line from Strategy 06.

QUESTION

| In triangle ABC , $AB = 9.4$ cm, $AC = 7.1$ cm and angle $A = 48^\circ$. Find BC .

YOUR SOLUTION

Quick Reference

TRIGGER → FIRST MOVE

WHEN YOU SEE	WRITE THIS FIRST
"work out", "show", "hence", a familiar exam phrase	Name the angle relationship
"work out", "show", "hence", a familiar exam phrase	Use the polygon sum
"work out", "show", "hence", a familiar exam phrase	Use exterior angles
"work out", "show", "hence", a familiar exam phrase	Chain reasons in order
"work out", "show", "hence", a familiar exam phrase	Use trigonometry
"work out", "show", "hence", a familiar exam phrase	Use cosine rule

FINAL BOTTOM LINE Do not try to remember every past-paper wording. Remember the trigger, write the first line, and let the working follow.